Worksheet 2

Measures of central tendency – EXERCISES

Badr ABOUFARASSE, PhD. – Academic year 2026–2027

Exercise 1 – Arithmetic, geometric, and harmonic means

1. A distribution takes values $x_i = 10, 20, 30$ with counts $n_i = 2, 3, 5$. Calculate the arithmetic mean.
2. Revenue changes by $+10\%$, -5%, and $+20\%$ in three successive years. Calculate the average annual growth rate using the geometric mean of growth factors.
3. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the average speed and explain why the harmonic mean is used.

Exercise 2 – Measures of central tendency for grouped data

Use the following willingness-to-pay distribution:

Price class (euros)	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
Count n_i	4	12	32	32	60	35	25	5	5

1. Identify the modal class, taking unequal class widths into account.
2. Calculate the median by interpolation.
3. Calculate Q_1 and Q_3 by interpolation.
4. Calculate the approximate arithmetic mean using class midpoints.
5. Interpret the modal class, median, mean, and quartiles.

Exercise 3 – Price affordable to 60% of customers

Find the maximum price such that about 60% of customers can afford the product. Use the cumulative curve or linear interpolation in the relevant class and explain the reasoning using cumulative frequencies.